

BACE1: why inhibitors failed, and whether PPI is the next bet

BACE1_Discovery canvas (print of all tabs)

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Peptides died on pharmacokinetics. Small molecules died after they worked: they lowered CSF amyloid 50-90% and still failed Phase 2/3, often with early cognitive worsening. A generic BACE1 PPI recreates that mechanism. An APP-selective interface strategy does not — if you can prove it.

Verdict. Do not pick a generic BACE1 protein-protein interaction as the primary thesis. Choose APP-BACE1 substrate-selective disruption only if you can spare SEZ6, NRG1, CHL1 and gp130, aim for ~30% A β lowering (Icelandic APP A673T range), and position the asset as prevention or combination with an anti-A β antibody — not as symptomatic monotherapy.

Metric	Value
Phase 3 BACE1 drugs, all stopped	5
Approved BACE1 inhibitors	0
CSF A β lowering with no clinical benefit	50-90%
Protective A β cut (APP A673T, in vitro ~40%)	~30% TPP band

1. Where they failed

Peptides / peptidomimetics (preclinical → Phase 1). Failed as drugs, not as enzyme blockers. OM99-2 (K_i ~1.6 nM) and KMI-429 were potent transition-state mimics: too large, hydrophilic, metabolically unstable, P-gp substrates, poorly brain-penetrant. CTS-21166 reached Phase 1 (2007-08), cut plasma A β ~80%, then never advanced.

Small-molecule active-site inhibitors (Phase 1 → Phase 3). Oral CNS delivery was solved. Late programs engaged the target, then failed from compound organ toxicity (Phase 1/2) and mechanism-based cognitive harm plus futility (Phase 2/3). Hair-color change (BACE2/PMEL), weight loss, psychiatric events, and MRI volume loss tracked across scaffolds — a class effect.

Clinical attrition

Asset	Class	Last stage	Population	Why it stopped	Primary sources
OM99-2	Octapeptide isostere	Preclinical	N/A	MW 893; no oral/BBB; CatD K _i 48 nM	Ghosh 2000; Hong 2000
KMI-429	Peptidomimetic	Preclinical	Mouse A β lowering	Acidic P4/P1'; poor BBB	Kimura 2004; Asai 2006
CTS-21166	Hydroxyethylamine	Phase 1 (2008)	Healthy volunteers	Plasma A β ~80%; no Phase 2	NCT00621010; Alzforum 2008
LY2811376	Small molecule	Phase 1 (2011)	Healthy / AD	Retinal toxicity (CatD)	May 2011; Zuhl 2016

Asset	Class	Last stage	Population	Why it stopped	Primary sources
LY2886721	Small molecule	Phase 2 (2013)	MCI / mild AD	Liver enzymes 4/70; Lilly: not BACE mechanism	May 2015; Lilly 2013 PR
RG7129	Small molecule	Phase 1 (2013)	Healthy	Roche stopped; liver toxicity reported (unofficial)	Alzforum; FierceBiotech 2014
Verubecestat EPOCH	Small molecule	Phase 3 (2017)	Mild-moderate AD	Futility; CSF A β -63 to -81%; early cognitive dip	Kennedy 2016; Egan 2018
Verubecestat APECS	Small molecule	Phase 3 (2018)	Prodromal AD, amyloid+	Futility + worse cognition and function	Egan 2019
Lanabecestat	Small molecule	Phase 3 (2018)	Early / mild AD	Futility; psych AEs, weight, hair, volume loss	Wessels 2020
Atabecestat EARLY	Small molecule	Phase 2b/3 (2018)	Preclinical amyloid+	Hepatotoxicity (11% ALT/AST >3 $\times$ ULN) + cognitive worsening	Henley 2019; Sperling 2021
Umibecestat Generation	Small molecule (BACE1-selective)	Phase 2/3 (2019)	APOE4 prevention	Early cognitive worsening; reversible after washout	Amgen 2019; Tariot 2025
Elenbecestat MISSION AD	Small molecule	Phase 3 (2019)	Early AD	DSMB: unfavorable risk-benefit; no efficacy signal	Eisai 2019; NCT02956486

Peptides never tested the amyloid hypothesis in patients. Small molecules did — and disproved high-level catalytic blockade as treatment for symptomatic or even prodromal AD.

2. Why biology killed them

Late BACE1 inhibitors were pharmacologically successful. CSF A β fell 50–90%. Clinical scores did not. In prodromal and prevention studies, cognition moved the wrong way within weeks to months — too fast to be failed amyloid clearance.

Approximate CSF / production A β reduction: APP A673T ~30–40% (lifetime, Jonsson 2012 ~40% in vitro); verubecestat 12 mg ~63%; 40 mg ~81% (Kennedy 2016 PD range); typical Phase 3 dose ~75%.

Substrate	Function at risk	Key source
SEZ6	Dendritic spines, LTP	Pigoni 2016; Pratsch 2023
CHL1 / L1	Axon guidance, plasticity	Kuhn 2012
NRG1	Myelination, ErbB	Willem 2006
gp130	Neuronal IL-6 buffering	Kuhn 2012
Jagged1	Notch-related	Kuhn 2012
BACE2 (off-target)	Hair pigment (PMEL)	Rochin 2013
Cathepsin D (off-target)	Retina / lysosome	Zuhl 2016

Three failure layers. (1) Too much inhibition (70–90% BACE loss; spine loss tracks sSEZ6, not scaffold; umibecestat was BACE1-preferring and still worsened cognition). (2) Wrong job for the stage (production brake does not clear plaques; antibodies do). (3) Compound toxicity on top (liver, retina/CatD, skin).

Umibecestat cognitive dip largely reversed after washout (Tariot 2025). EPOCH week-13 ADAS-Cog dip (Egan 2018).

3. PPI decision

Only if the PPI is the APP-BACE1 recognition interface, as substrate-selective partial modulation. An exosite antibody that shuts BACE1 for every substrate is verubecestat biology with worse CNS delivery.

Do this: APP-selective BACE1 inhibition (ASBI). FAH65 / FAH65E(–) (Campagna 2025): APP IC50 far below NRG1/PSGL1; little CatD/BACE2; brain exposure; NOR in mice. Still preclinical. Design goal 20–40% A β , CSF SEZ6/sNRG1 unchanged, combo with anti-A β mAb.

Do not do this: Generic BACE1 PPI / catalytic exosite block. Atwal 2011 / PDB 3R1G: BACE2/CatD-selective but still inhibits BACE1 toward APP. BBB needed a TfR shuttle (Yu 2011), later used for A β antibodies, not a clinical BACE1 drug.

Go/no-go: substrate panel must spare SEZ6/NRG1/CHL1/gp130; inhibition ceiling 20–50% in vivo without spine/LTP loss; oral CNS or real BBB shuttle; prevention or mAb add-on, not monotherapy in mild-moderate AD.

References

1. Ghosh AK, et al. *J Am Chem Soc.* 2000;122:3522–3523. doi:10.1021/ja000300g
2. Hong L, et al. *Science.* 2000;290:150–153. doi:10.1126/science.290.5489.150
3. Kimura T, et al. *Bioorg Med Chem Lett.* 2005;15:211–215. doi:10.1016/j.bmcl.2004.09.090
4. Asai M, et al. *J Neurochem.* 2006;96:533–540. doi:10.1111/j.1471-4159.2005.03576.x
5. NCT00621010; Alzforum Keystone 2008 (CTS-21166, plasma A β up to ~80%).
6. May PC, et al. *J Neurosci.* 2011;31:16507–16516. doi:10.1523/JNEUROSCI.3647-11.2011
7. Zuhl AM, et al. *Nat Commun.* 2016;7:13042. doi:10.1038/ncomms13042
8. May PC, et al. *J Neurosci.* 2015;35:1199–1210. doi:10.1523/JNEUROSCI.4129-14.2015
9. Eli Lilly press release, 13 Jun 2013 (LY2886721; liver tests not attributed to BACE).
10. Alzforum RG7129; FierceBiotech 2014. Roche issued no official liver-toxicity CSR.
11. Kennedy ME, et al. *Sci Transl Med.* 2016;8:363ra150. doi:10.1126/scitranslmed.aad9704
12. Egan MF, et al. *N Engl J Med.* 2018;378:1691–1703. doi:10.1056/NEJMoa1706441
13. Egan MF, et al. *N Engl J Med.* 2019;380:1408–1420. doi:10.1056/NEJMoa1812840
14. Wessels AM, et al. *JAMA Neurol.* 2020;77:199–209. doi:10.1001/jamaneurol.2019.3988
15. Henley D, et al. *N Engl J Med.* 2019;380:1483–1485. doi:10.1056/NEJMc1813435
16. Sperling R, et al. *JAMA Neurol.* 2021;78:293–301. doi:10.1001/jamaneurol.2020.4857
17. Amgen/Novartis/Banner PR, 11 Jul 2019 (umibecestat / CNP520).
18. Tariot PN, et al. *Alzheimers Dement.* 2025;21:e14237. doi:10.1002/alz.14237

19. Eisai/Biogen, 13 Sep 2019; [NCT02956486](#)
20. Jonsson T, et al. *Nature*. 2012;488:96–99. doi:[10.1038/nature11283](#)
21. Piloni M, et al. *Mol Neurodegener*. 2016;11:67. doi:[10.1186/s13024-016-0134-z](#)
22. Pratsch K, et al. *Int J Mol Sci*. 2023;24:12283. doi:[10.3390/ijms241512283](#)
23. Kuhn PH, et al. *EMBO J*. 2012;31:3157–3168. doi:[10.1038/emboj.2012.132](#)
24. Willem M, et al. *Science*. 2006;314:664–666. doi:[10.1126/science.1132341](#)
25. Rochin L, et al. *Proc Natl Acad Sci USA*. 2013;110:10658–10663. doi:[10.1073/pnas.1221294110](#)
26. van Dyck CH, et al. *N Engl J Med*. 2023;388:9–21. doi:[10.1056/NEJMoa2212948](#)
27. Sims JR, et al. *JAMA*. 2023;330:512–527. doi:[10.1001/jama.2023.13239](#)
28. Campagna J, et al. *Neurotherapeutics*. 2025;e00610. doi:[10.1016/j.neurot.2025.e00610](#)
29. Atwal JK, et al. *Sci Transl Med*. 2011;3:84ra43. doi:[10.1126/scitranslmed.3002254](#)
30. Yu YJ, et al. *Sci Transl Med*. 2011;3:84ra44. doi:[10.1126/scitranslmed.3002230](#)